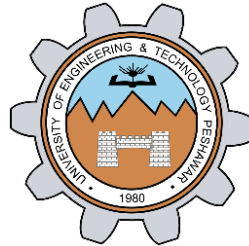


LAB # 02
INTERFACING LEDS TO 8051 AND DISPLAYING BINARY PATTERNS



Spring 2024
CSE-307L Microprocessor Based System Design Lab

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“On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

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Objectives of The Lab:

- Interfacing leds with 8051 microcontroller
- New assembly instructions ACALL, RET DJNZ
- Coding loop and nested loop in assembly

Lab Tasks:

Display the pattern using assembly and C language.

```
00000001
00000010
00000100
....
10000000
00000001
00000010
```

Assembly Code:

```
1  ORG 0000H
2  HERE:
3  MOV P0,#01H ;0000 0001
4  ACALL DELAY
5  MOV P0,#02H ;0000 0010
6  ACALL DELAY
7  MOV P0,#04H ;0000 0100
8  ACALL DELAY
9  MOV P0,#08H ;0000 1000
10 ACALL DELAY
11 MOV P0,#10H ;0001 0000
12 ACALL DELAY
13 MOV P0,#20H ;0010 0000
14 ACALL DELAY
15 MOV P0,#40H ;0100 0000
16 ACALL DELAY
17 MOV P0,#80H ;1000 0000
18 ACALL DELAY
19 MOV P0,#40H ;0100 0000
```

```
19 MOV P0,#40H ;0100 0000
20 ACALL DELAY
21 MOV P0,#20H ;0010 0000
22 ACALL DELAY
23 MOV P0,#10H ;0001 0000
24 ACALL DELAY
25 MOV P0,#08H ;0000 1000
26 ACALL DELAY
27 MOV P0,#04H ;0000 0100
28 ACALL DELAY
29 MOV P0,#02H ;0000 0010
30 ACALL DELAY
31 SJMP HERE
```

```
33 DELAY:
34 MOV R0,#5
35 MOV R1,#5
36 MOV R3,#10
37 LOOP1:
38     LOOP2:
39         LOOP3:
40             DJNZ R3,LOOP3
41             DJNZ R1,LOOP2
42             DJNZ R0,LOOP1
43     RET
44 END
```

C Code:

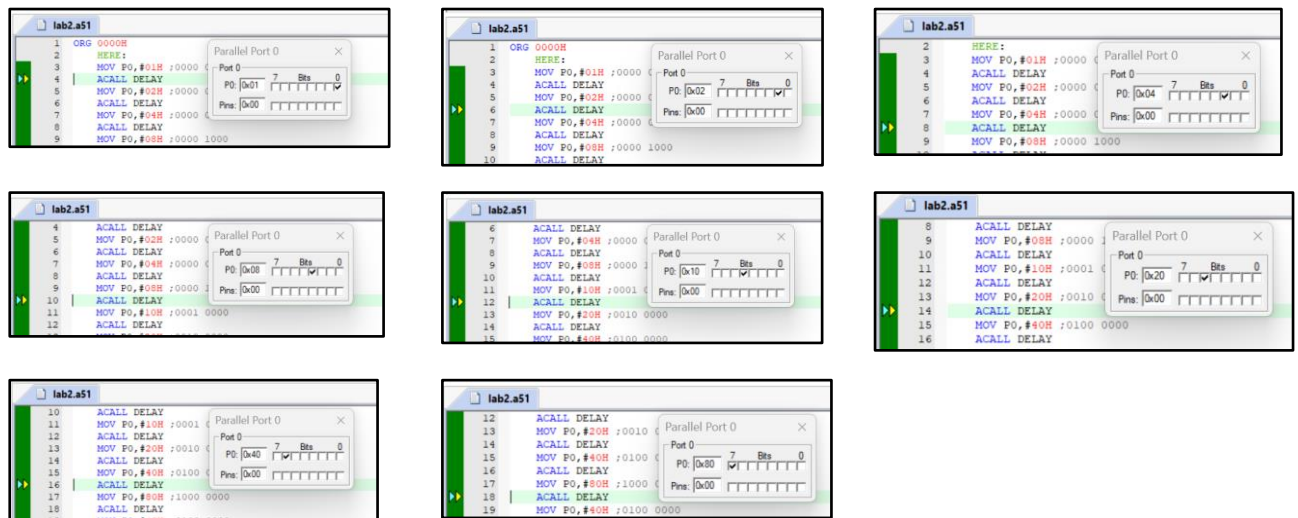
```
1 #include <reg51.h>
2 int A;
3
4 void delay(){
5     for(A=0; A<5; A++){
6         for(A=0; A<5; A++){
7             for(A=0; A<10; A++){
8             }
9         }
10    }
11 }
12 return;
13 }
```

```
14 void main(){
15     while(1){
16         A = 0x01;
17         P0 = A;
18         delay();
19         A = 0x02;
20         P0 = A;
21         delay();
22         A = 0x04;
23         P0 = A;
24         delay();
25         A = 0x08;
26         P0 = A;
27         delay();
28         A = 0x10;
29         P0 = A;
30         delay();
31         A = 0x20;
32         P0 = A;
```

```
33         delay();
34         A = 0x40;
35         P0 = A;
36         delay();
37         A = 0x80;
38         P0 = A;
39         delay();
40         A = 0x40;
41         P0 = A;
42         delay();
43         A = 0x20;
44         P0 = A;
45         delay();
46         A = 0x10;
47         P0 = A;
48         delay();
49         A = 0x08;
50         P0 = A;
51         delay();
52         A = 0x04;
```

```
53         P0 = A;
54         delay();
55         A = 0x02;
56         P0 = A;
57         delay();
58     }
59 }
```

Simulation in Keil:



Simulation in Proteus:

The following pictures show the simulation of leds in proteus.

